

Causal-augmented disentanglement for contrastive recommendation

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Graph neural networks (GNNs) combined with contrastive learning have shown promise in mitigating data sparsity and noise in recommender systems. However, current approaches face two key challenges: (1) they often conflate genuine user preferences with group-level influences, introducing confounding bias in representation learning; (2) conventional graph contrastive methods rely on random perturbations that can compromise semantic structure and amplify noise. To overcome these issues, we propose CaDCR — Causal-Augmented Disentanglement for Contrastive Recommendation. CaDCR introduces a dual-channel mechanism to separate individual and collective intent, applying causal interventions to eliminate bias from factors like popularity. Additionally, it employs dual contrastive regularization in both interaction and intent spaces, preserving semantic integrity and enhancing representation quality without manual augmentation. Experiments on Gowalla and other datasets show CaDCR outperforms 14 mainstream methods on metrics like Recall and NDCG, validating its effectiveness and practical relevance.

Keywords: Intent disentanglement; graph neural networks; contrastive learning; causal inference; recommender systems.

1. Introduction

1.1. Background

Recommender systems are fundamental to mitigating information overload, enabling personalized content delivery by modeling user behavior and item attributes are a key technology for tackling information overload, generating personalized recommendations for different users by analyzing user behavior and item characteristics, with important value in news, social networks and other domains.¹ Collaborative filtering, which predicts interests by uncovering latent associations between users and items, laid the foundation for scalable personalized recommendation.² In recent years, graph-based recommendation models have gradually gained prominence.³ Graph neural networks (GNNs) explicitly enhance node representations by iteratively aggregating neighbor information and exploiting high-order connectivity, as exemplified by approaches such as PinSage⁴ and GraphSAGE,⁵ which have significantly improved recommendation performance, especially in long-tail scenarios. Nevertheless, such methods still face two core challenges. On the one hand, the sparsity of interaction data and the noise inherent in implicit feedback can severely impair model generalization. On the other hand, traditional approaches often simplistically attribute observed interactions to users' stable preferences, failing to adequately model the complexity of user decision-making

motives — where genuine interests, conformity psychology, item popularity and other factors are entangled within a single representation. This makes it difficult to accurately capture users’ true intents and effectively counteract the interference of confounders such as popularity in representation learning, ultimately leading to biased recommendation outcomes.^{6,7} To address these limitations, contrastive learning mechanisms have been introduced into recommendation frameworks: these approaches, in fact, enable the extraction of discriminative features via carefully constructed positive and negative samples, without requiring labeled data and they lead to relevant performance gains. Concurrently, causality-based recommendation methods are gaining attention; specifically, these methods attempt to bring causal mechanisms into recommender systems to address biases and fairness issues in traditional models.⁸ Causal disentangled techniques, in particular, aim to decompose user interactions into representations corresponding to different causal factors, and, thus, they provide a new perspective on understanding user decision processes. For example, DICE⁹ decomposes user interactions into independent components of interest and conformity, using causal inference’s counterfactual framework to learn representations for these two parts; CausE¹⁰ proposes a domain-adaptive random exposure result prediction algorithm for causal recommendation and strategy optimization under biased logs; DecRS¹¹ employs causal front-door adjustments with approximate operators and dynamic reasoning to eliminate sample distribution confounding on predicted scores.

1.2. Research motivation

Although many studies have advanced in disentangling intents, two core challenges remain for most recommender methods: (i) inability to effectively distinguish intrinsic user preferences from external population effects. Such an inability leads to substantial confounding in representation learning. This bias stems from the “confounding triangle” structure — collective intent: in detail, popularity acts as a confounder influencing both user decisions and item exposure, making it hard to tell whether a click is due to true interest or herd behavior. The result is over-promotion of popular items and under-exposure of the long tail. (ii) Most existing contrastive learning approaches rely on heuristic data augmentation such as random perturbations; such randomization can disrupt semantic correlations in the graph structure, leading to loss of key cooperative signals. This compromises structural integrity, introduces training uncertainty and limits discriminability and generalization. To address these issues, we propose two key innovations:

- (1) Causal Modeling of User Decisions: We decompose interactions into intrinsic preferences and collective trends, enabling a dual-channel framework that explicitly separates confounded signals. Our solution improves robustness under sparse data and dynamic environments by isolating true user intent.
- (2) Dual-Space Contrastive Regularization: We design a structural contrastive task to preserve graph semantics while enforcing distribution uniformity in the feature space. Joint alignment across structure and features stabilizes self-supervised

learning and it simultaneously enhances generalization, yielding more robust user and item representations.

1.3. Main contributions

The main contributions of this paper are summarized as follows:

- (1) We introduce a novel mechanism that disentangles individual and collective intents and applies causal intervention to remove confounding effects such as popularity. This yields unbiased user and item representations that better capture genuine interests and mitigate popularity bias.
- (2) We design a dual-space graph contrastive regularization (GCR) method Causal-augmented disentanglement for contrastive recommendation (CaDCR) operating on interaction and intent spaces. Unlike heuristic augmentations, CaDCR constructs structural contrastive tasks in the interaction topology and enforces distributional uniformity in the feature space. This preserves graph semantics, enhances discriminability and delivers stable, comprehensive representations without sacrificing structural integrity.
- (3) We tested CaDCR on three popular real-life datasets, namely Amazon-Book, Gowalla and Tmall. Our experiments suggest that CaDCR consistently outperforms 14 strong baselines — including LightGCN and BIGCF — across all metrics, demonstrating superior accuracy and robustness.

2. Related Work

2.1. Recommender systems based on GNNs

The core of recommender systems is to learn latent preferences from the user-item interaction history. Since interaction data can naturally be represented as a bipartite graph, GNNs have become a central architecture in modern recommender systems due to their powerful capability to model structural information.^{12–14} Early representative works include NGCF,¹⁵ which propagates embeddings on the graph to capture higher-order connectivity and explicitly injects collaborative filtering signals into the embedding process. LightGCN,¹⁶ which simplifies the architecture of a standard Graph Convolution Network (GCN) by preserving only the core neighborhood aggregation component and propagates embeddings linearly on the interaction graph, yields a lightweight and more effective architecture. To capture more complex user-item relationships, LADIES¹⁷ constructs a bidirectional adjacency graph; on the upper layers, it selects neighboring nodes, and builds a full computation graph by performing fixed-sampling with a neighborhood importance probability. Cluster-GCN¹⁸ uses graph clustering to select compact subgraphs, localizing SGD training to reduce memory and computation overhead. FastGCN¹⁹ treats GCN as an integral transform of the embedding function, introducing importance sampling to improve training and inference generalization; SHADOW-GNN²⁰ extracts a

Table 1. Recommender systems based on GNN.

Authors	Main idea	Year
Wang <i>et al.</i> ¹⁵	Propagate embeddings on the graph to capture higher-order connectivity, explicitly injecting collaborative filtering signals into the embedding process.	2019
He <i>et al.</i> ¹⁶	The architecture of a standard Graph Convolutional Network (GCN) by retaining only the core neighborhood aggregation component and linearly propagating embeddings on the interaction graph.	2020
Zou <i>et al.</i> ¹⁷	Select neighboring nodes at higher layers and construct a complete computation graph through fixed sampling with neighborhood importance probability	2019
Chiang <i>et al.</i> ¹⁸	Use graph clustering to select compact subgraphs and localize SGD training	2019
Chen <i>et al.</i> ¹⁹	Treat GCN as an integral transform of the embedding function and introduce importance sampling	
Zeng <i>et al.</i> ²⁰	Extract a finite set of key neighbors within a local subgraph and apply arbitrarily deep GNNs on this subgraph to represent the target node/edge.	2021

finite set of key neighbors within a local subgraph and applies arbitrary-depth GNNs on this subgraph to represent the target node/edge. A summary based on Table 1 is provided for the above content.

2.2. Recommender systems based on contrastive learning

Contrastive learning^{21,22} has achieved remarkable success in computer vision, natural language processing and other fields in recent years. Its core idea is to bring representations of semantically similar samples closer and push apart representations of dissimilar samples. In recommender systems, methods are able to learn high-quality, robust user and item representations through their strong self-supervised signals. For example, GRACE²³ generates mixed views at the graph structure and attribute levels to maximize consistency of node representations across two views. GAUG²⁴ augments the existing graph by edge prediction to generate enhanced information, improving GNN node classification performance and thereby enabling a graph augmentation framework. GraphCL²⁵ obtains graph representations through four graph augmentation strategies to achieve a self-supervised graph contrastive learning framework. XSimGCL²⁶ uses simple noise-injection augmentation to replace traditional graph augmentations, and, thus, it improves the uniformity of user/item representations and alleviates cold-start and long-tail biases. NCL²⁷ introduces structural and semantic neighbors into graph contrastive learning for collaborative filtering, using structure contrast and prototype contrast to boost robustness and recommendation performance. GGDHSCl²⁸ designs a dual-channel view generator to establish dual-view contrast, alleviating data sparsity and noise issues and enhances the self-supervised signal via hard negative sampling. In Table 2, we present a summary of several state-of-the-art contrastive recommendation algorithms.

Table 2. Recommender systems based on GNN.

Authors	Main idea	Year
Zhao et al. ²⁴	Augments existing graphs via edge prediction to generate enhanced structural information.	2023
You et al. ²⁵	Leverages four graph augmentation strategies to obtain robust graph representations, forming a self-supervised graph contrastive learning framework.	2020
Yu et al. ²⁶	Employs simple noise-injection augmentation instead of traditional graph augmentations.	2024
Lin et al. ²⁷	Introduces structural and semantic neighbors into graph contrastive learning, using structure and prototype contrast to enhance robustness and recommendation performance.	2022
Liu et al. ²⁸	Designs a dual-channel view generator for dual-view contrastive learning, alleviating data sparsity and noise issues through hard negative sampling.	2025

2.3. Recommender systems based on disentanglement

A key focus in modern recommender systems is disentangling explainable latent motives from mixed user behavior. Disentangled representation methods address this issue by decomposing user and item embeddings into independent, semantically clear components, each of which captures specific intentions, such as brand preference or seasonal demand. This enables fine-grained modeling of complex decision-making processes. For example, GNUD²⁹ introduces neighborhood routing to decouple user preference factors and subspace representations, with preference regularization encouraging each subspace to independently reflect different preferences. GD^{CF}³⁰ maps user-item historical interactions into multiple geometric spaces, learning geometric disentangled representations and a generative model of high-level intent concepts to reveal latent intent factors. HIDE³¹ constructs a hypergraph for each session to capture multi-view interest transitions, performing intent-aware embedding propagation at the micro-level for each click to suppress noise and adding auxiliary intent classification tasks at the macro level to promote independence among different intents. MIDGN³² decouples intent globally across bundled items while decoupling within-bundle item relationships locally. MCPRN³³ allocates items in a session to different channels by purpose, and builds purpose-specific recurrent networks for each channel to capture item dependencies under the same goal; MITGNN³⁴ uses tail entity translation and relation vector learning, coupled with multi-head aggregation to process user/item information; ICM-SR³⁵ generates an initial intent representation from the preceding items and the last item of the current session, then selects relevant neighboring sessions via an intent-guided neighbor detector, and finally integrates session and neighbor representations through gated fusion.

However, it is worth noting that most existing methods have not distinguished intrinsic user preferences from external group influences from a causal perspective, making it difficult to accurately capture users' true intentions. Additionally, the contrastive learning approaches used can disrupt the semantic integrity of graph structure, which limits the robustness and generalization of the models.

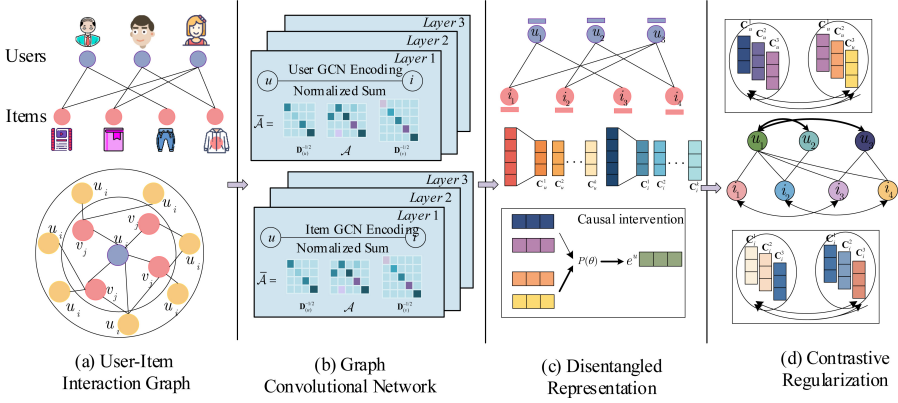


Fig. 1. System architecture diagram of the CaDCR system.

3. Proposed Methodology

The CaDCR framework consists of four core modules: (1) the interaction graph construction module transforms data into graph structures; (2) the high-order GNN encoder generates high-order neighborhood representations through message propagation; (3) the causal intent disentanglement module separates genuine user preferences from confounding factors based on causal inference; (4) the multi-task curriculum learning module learns uniform and clear disentangled representations via progressive tasks and contrastive regularization, thereby enhancing generalization capability. The design of each module will be elaborated in detail below (Fig. 1).

3.1. Problem definition

In the following, we denote the set of all users is as $\mathcal{U} = \{u_1, u_2, \dots, u_M\}$, and the set of all items as $\mathcal{I} = \{i_1, i_2, \dots, i_N\}$, where M and N represent the total number of users and items, respectively. Based on the historical interactions between users and items, a user-item interaction matrix can be constructed. In this matrix, if there exists some form of interaction between user u and item i (e.g. click, purchase, or rating), the corresponding entry is $\mathcal{A}_{u,i} = 1$; otherwise, $\mathcal{A}_{u,i} = 0$.

The interaction matrix is often represented as a bipartite graph, denoted as $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$, where the node set $\mathcal{V} = \{\mathcal{U} \cup \mathcal{I}\}$ includes all user and item nodes and the edge set $\mathcal{E} = \{(u, i) | u \in \mathcal{U}, i \in \mathcal{I}, \mathcal{A}_{u,i} = 1\}$ records observed interactions.

On this basis, the probability that a target user u forms a new connection with an item i with which they have not yet interacted in the interaction graph $\mathcal{G}$ can be modeled as the following conditional probability form:

$$\hat{y}_{ui} = (\hat{\mathcal{A}} | \mathbf{E}^{(0)}, \mathcal{A}) = \prod_{u \in \mathcal{U}} \prod_{i \in \mathcal{I}} (\hat{\mathcal{A}}_{ui} | \mathbf{e}_u, \mathbf{e}_i), \quad (1)$$

where each user u and item i have initial embedding vectors denoted as $\mathbf{E}^{(0)} = \{\mathbf{E}_U^{(0)}, \mathbf{E}_I^{(0)}\}$. $\mathbf{e}_u$ and $\mathbf{e}_i$ are the embeddings of user u and item i after encoding.

All user and item embedding vectors together form the embedding sets $E_u \in M \times d$ and $E_i \in J \times d$, where d represents the embedding dimension.

Furthermore, since user-item interactions are often driven by multiple latent intent factors, these factors form a complex motivational background in the user's decision process. For example, some users may choose an item due to a preference for a specific color (e.g. purple), while others may be mainly influenced by the item's popularity. To more precisely characterize this multi-motive decision mechanism, we explicitly introduce a set of latent intent factors $\mathbf{C}$.

$$\begin{aligned}\hat{y}_{ui} &= (\hat{\mathcal{A}}|\mathbf{E}^{(0)}, \mathcal{A}, \mathbf{C}) \\ &= \prod_{u \in \mathcal{U}} \prod_{i \in \mathcal{I}} (\hat{\mathcal{A}}_{ui} | \mathbf{e}_u, \mathbf{e}_i) \times \sum_{k \in \mathcal{K}} (\mathbf{e}_u | \mathbf{E}^{(0)}, \mathbf{R}, \mathbf{C}_U^k) (\mathbf{e}_i | \mathbf{E}^{(0)}, \mathcal{A}, \mathbf{C}_I^k),\end{aligned}\quad (2)$$

where $\mathbf{C} = \{\mathbf{C}_U^k, \mathbf{C}_I^k | k \in \mathcal{K}\}$ represents K different intent types, which from a macro perspective jointly characterize the common preference patterns of user groups and the universal driving forces behind item selection. Specifically, user intent $\mathbf{C}_U$ reflects the group's convergence tendency or herd behavior, and it captures the decision motives shared by most users when making choices; correspondingly, item intent $\mathbf{C}_I$ corresponds to the popularity effect of the item, describing its inherent attractive attributes that make it broadly acceptable to a wide range of users. Integrating these latent intent factors explicitly into the probabilistic generation framework enables the model to better understand and simulate the complex, multi-factor decision-making processes that underlie user-item interactions.

3.2. Higher-order collaborative signal encoding

A large body of existing research has provided ample empirical evidence to validate the effectiveness of using a GNN as the model encoder. Building on this consensus, this paper adopts a lightweight Graph Convolutional Network (LightGCN) as its core architecture to efficiently encode high-order collaborative relationships between users and items. The specific encoding process is as follows:

Within the GCN modeling framework, the user-item interaction data are modeled as a bipartite graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$. Subsequently, the model aggregates information from neighbor nodes through a multi-layer embedding propagation mechanism, with the propagation function defined at each layer as follows:

$$\begin{aligned}\mathbf{e}_u^{(l+1)} &= \sum_{i \in \mathcal{N}_u} \frac{1}{\sqrt{|\mathcal{N}_u|} \sqrt{|\mathcal{N}_i|}} \mathbf{e}_i^{(l)}, \\ \mathbf{e}_i^{(l+1)} &= \sum_{u \in \mathcal{N}_i} \frac{1}{\sqrt{|\mathcal{N}_i|} \sqrt{|\mathcal{N}_u|}} \mathbf{e}_u^{(l)}.\end{aligned}\quad (3)$$

Let $\mathbf{e}_u^{(l)}, \mathbf{e}_i^{(l)}$ denote the embeddings of user u and item i at the l th layer, respectively. Let $\mathcal{N}_u$ and $\mathcal{N}_i$ be the sets of first-order neighbor nodes for user u and item i , respectively. The coefficient $1/\sqrt{|\mathcal{N}_u|} \sqrt{|\mathcal{N}_i|}$ is the unified normalization term in the

GNN framework, used to balance the impact of node degree differences on information propagation. After L layers of embedding propagation, the final graph-structured embeddings for users and items are obtained by averaging the representations from all layers, including the initial layer (0th layer):

$$\mathbf{e}_u = \frac{1}{L+1} \sum_{i=0}^L \mathbf{e}_u^{(i)}, \quad \mathbf{e}_i = \frac{1}{L+1} \sum_{i=0}^L \mathbf{e}_i^{(i)}, \quad (4)$$

where $\mathbf{e}_u$ and $\mathbf{e}_i$ denote the final embeddings of user u and item i on the graph, respectively, which integrate multi-layer collaborative signals from first-order to higher-order, providing a structurally enriched representation foundation for downstream preference prediction tasks.

3.3. Disentangled representations

Inspired by the latest advances in GNNs, this paper designs a GNN-based disentangled representation learning module aimed at learning distributed representations from the user-item interaction graph that can intrinsically reveal K different user intents. Specifically, an L -layer disentangled module restructures the initial embeddings of users and items by leveraging higher-order connectivity in the graph, dividing them into K independent embedding blocks, thereby initializing intent-aware embedding representations. On this basis, an interactive update rule is designed to dynamically compute the importance scores of user-item interactions along each intent dimension, thereby finely adjusting the embeddings corresponding to each intent. Subsequently, through embedding propagation at the current network layer, information is aggregated and state-updated for each intent-aware embedding block, generating the overall intent-aware embeddings for users u and items i , respectively. The intents of users and items can be formalized in a concrete form as

$$\mathbf{C}_u^k = [\mathbf{C}_u^1, \dots, \mathbf{C}_u^k], \quad \mathbf{C}_i^k = [\mathbf{C}_i^1, \dots, \mathbf{C}_i^k]. \quad (5)$$

Perform embedding propagation at each layer using the relevance score for each intent k :

$$\begin{aligned} \mathbf{i}_{i,l}^{(u)} &= \sum_k^K \mathbf{c}_u^i \frac{\exp(\mathbf{e}_{i,l-1}^{(u)\top} \mathbf{C}_u^k)}{\sum_{h'}^K \exp(\mathbf{e}_{i,l-1}^{(u)\top} \mathbf{C}_u^{k'})}, \\ \mathbf{i}_{j,l}^{(v)} &= \sum_k^K \mathbf{c}_v^j \frac{\exp(\mathbf{e}_{j,l-1}^{(v)\top} \mathbf{C}_v^k)}{\sum_{h'}^K \exp(\mathbf{e}_{j,l-1}^{(v)\top} \mathbf{C}_v^{k'})}. \end{aligned} \quad (6)$$

The final disentangled representations for users and items can be constructed as the combined output from each layer:

$$\begin{aligned} \mathbf{e}_u &= \mathbf{e}_u^{(l+1)} + \mathbf{i}_{i,l}^{(u)}, \\ \mathbf{e}_i &= \mathbf{e}_i^{(l+1)} + \mathbf{i}_{j,l}^{(v)}. \end{aligned} \quad (7)$$

3.4. Causal intervention

Traditional recommendation methods primarily use correlations to learn, which makes it difficult to distinguish whether a user's click behavior stems from real personal interest or herd behavior influenced by group effects. Contextual information acting as a confounder can introduce systematic biases into the learning of user representations and the final predicted scores. To improve the identification of intent at the semantic level, this paper introduces causal inference techniques to perform causal interventions on user representations. The aim is to address the confounding of individual intent and contextual factors in observational data and obtain unbiased user representations.

Specifically, if we ignore the potential causal structure in user-item interactions, the model's predicted behavior would be equivalent to evaluating $P(Y|U)$, where U denotes the user representation, which essentially reflects latent intent (for example, preferences for specific brands or colors) and $Y \in [0, 1]$ represents the probability of a user-item pair being recommended.

To reframe the problem from a causal perspective, the prediction target is represented as the distribution after intervention, $P(Y|\text{do}(U = u))$ introduced $\text{do}(\cdot)$ as the core mechanism of causal intervention.

After performing the $\text{do}(\cdot)$ operation, the model can be further expressed as

$$P(Y|\text{do}(U = u)) - P(Y|\text{do}(U = 0)) = \frac{1}{N} \sum_{i=1}^N (P(\hat{y}_{ui}|i \odot H) - P(\hat{y}_{ui}|i)), \quad (8)$$

where $P(Y|\text{do}(U = 0))$ represents zero intervention, in the real world $P(Y|\text{do}(U = u))$ corresponds to aspects or attributes in the active intervention terms, and H denotes confounders; these $\hat{y}_C$ and $\hat{y}$ are computed through $\hat{P}(Y|\text{do}(\hat{y}_{ui} = i \odot H))$ and $P(Y|\text{do}(\hat{y}_{ui} = 0))$ to define a function $\mathbf{D}_a$ that considers whether $H = h_a$ to incorporate them into the user's intent:

$$\mathbf{D}_a := \begin{cases} h_a & \text{if } \tanh(\hat{y}_C - \hat{y}) > 0, \\ 1 & \text{if } \tanh(\hat{y}_C - \hat{y}) < 0, \end{cases} \quad (9)$$

where $\tanh(\hat{y}_C - \hat{y}) > 0$ indicates factors that would positively influence the recommendation result, otherwise would have a negative influence and should be removed from the user intent representation. Consequently, the user intent representation can be further expressed as

$$E_a = e_u \odot \mathbf{D}_a. \quad (10)$$

3.5. Graph contrastive learning

Although Graph Convolutional Networks can learn disentangled intent representations from the raw user-item interaction graph, this learning process still faces challenges from sparse connections and noisy interactions. A large body of research

shows that contrastive learning can effectively alleviate over-smoothing and noise interference in GNNs. Given an arbitrary node a in the graph, its contrastive loss can be expressed as

$$I(\mathbf{a}', \mathbf{a}'') = -\log \frac{\exp(\cos(\mathbf{a}', \mathbf{a}'')/\tau)}{\sum_{b \in \mathcal{B}} \exp(\cos(\mathbf{a}', \mathbf{b})/\tau)}, \quad (11)$$

where $\mathbf{a}'$ and $\mathbf{a}''$ are two augmented embeddings of node a obtained through data augmentation, $\cos(\cdot)$ denotes cosine similarity, and τ is a predefined temperature coefficient.

Other nodes $\{b \in \mathcal{B}\}$ in the mini-batch are treated as negative samples for node a . However, most existing graph data augmentation techniques not only significantly increase computational overhead but also risk losing critical structural information — the augmentation operations may disrupt the topology of the original interaction graph, thereby degrading the semantic integrity of the input data.

At the same time, the core mechanism of InfoNCE tends to encourage a uniform distribution in the feature space.

To address this issue, the contrastive learning process is redesigned, treating it as $T(\mathbf{a}, \mathbf{b})$, a structured regularization operation over all nodes $\mathcal{V}$ in the interaction graph $\mathcal{G}$, with the main steps as follows:

- (1) Contrastive interaction regularization $T(\mathbf{e}_u, \mathbf{e}_i)$. The essence of the recommendation task is to measure semantic similarity between users and items, requiring that user u and its positive interaction item i be as close as possible in embedding space, while distancing from negative samples $\{j \in \mathcal{B}/i\}$.
- (2) Neighborhood-consistent regularization $T(\mathbf{e}_u, \mathbf{e}_u)$ and $T(\mathbf{e}_i, \mathbf{e}_i)$. To prevent learned embeddings from collapsing in the feature space, treat the node itself as a positive sample and other distant nodes in the graph as negative samples. The GCR in the interaction space can be formalized as

$$\mathcal{L}_{cl} = \sum_{\langle u, i \rangle \in \mathcal{B}} (T(\mathbf{e}_u, \mathbf{e}_i) + T(\mathbf{e}_u, \mathbf{e}_u) + T(\mathbf{e}_i, \mathbf{e}_i)), \quad (12)$$

where $\mathbf{e}_u$ and $\mathbf{e}_i$ are the user and item embeddings obtained via Graph Convolutional Networks. This interaction regularizer reduces computational complexity and preserves the integrity of the graph structure, while ensuring feature alignment between positive interaction pairs and consistency across the entire interaction space. In comparison, traditional contrastive learning strategies often overlook the bidirectional alignment between users and items.

- (3) Bilateral intent regularization $T(\mathbf{i}_{i,l}^{(u)}, \mathbf{i}_{i,l}^{(u)})$ and $T(\mathbf{i}_{j,l}^{(v)}, \mathbf{i}_{j,l}^{(v)})$: by learning individual intents $\{\mathbf{i}_{i,l}^{(u)}, \mathbf{i}_{j,l}^{(v)}\}$ in the intent space, and collective intents $\{\mathbf{C}_U, \mathbf{C}_I\}$. Considering that these intent embeddings reflect diverse motivations for a user to choose an item (or for an item to be chosen), such that an intent k could be represented as a linear combination of other intents $\{k' \in K/k\}$, making

it potentially redundant. Based on this, we further constrain the disentangled intent representations through GCR:

$$\mathcal{L}_{it} = \sum_{u \in \mathcal{B}} T(\mathbf{i}_{i,l}^{(u)}, \mathbf{i}_{i,l}^{(u)}) + \sum_{i \in \mathcal{B}} T(\mathbf{i}_{j,l}^{(v)}, \mathbf{i}_{j,l}^{(v)}). \quad (13)$$

This regularization term encourages independence and diversity between different intent dimensions, avoiding redundancy and overlap in intent representations.

3.6. Model training

During the training phase, the classic Bayesian Personalized Ranking (BPR) framework³⁶ is used to optimize recommendation performance. Its core objective is to rank items preferred by the user higher than items not observed by the user:

$$\mathcal{L}_{\text{BPR}} = \sum_{(u,i,j) \in O} -\log \sigma(y_{u,i} - y_{u,j}), \quad (14)$$

where $\sigma(\cdot)$ represents the sigmoid activation function, $O = (u, i, j) | (u, i) \in O^+, (u, j) \in O^-, O^+$ denotes observed user-item interactions, and O^- denotes sampled items that the user has never interacted with. Finally, the self-supervised contrastive loss and the BPR loss are integrated into a multi-task learning framework:

$$\mathcal{L} = \mathcal{L}_{\text{BPR}} + \lambda_1(\mathcal{L}_d + \mathcal{L}_{it}) + \lambda_2\Theta^2, \quad (15)$$

where λ_1, λ_2 are the tunable weights. The CaDCR algorithm is presented as follows:

Algorithm 1. Learning Process of CaDCR

Input: User-item interaction graph $\mathcal{G}$, initial embeddings, hyperparameters L, K

Output: Final refined user/item representations.

1. Initialize user and item embeddings;
 2. Initialize $\mathbf{e}_u$ and $\mathbf{e}_i$
 3. for epoch=0 to max epoch do
 4. for l to L do
 5. Propagate embeddings via LightGCN neighbor aggregation;
 6. Normalize aggregated embeddings;
 7. Compute final embeddings as mean of all layer representations.;
 8. Apply causal intervention to remove confounding bias;
 9. Concatenate adjusted intent as disentangled representation.
 10. Combine regularization losses and update model parameters;
 11. end;
 12. Update the parameters of the GNN
 13. end for;
-

4. Experiments and Analysis

We conducted comparative experiments on three public datasets—Gowalla, Tmall and Amazon-book — with 14 baseline methods, and address the following questions:

- (1) RQ1: How does the model perform compared with other mainstream top- p recommendations?
- (2) RQ2: How do the proposed key components affect recommendation performance?
- (3) RQ3: Compared with other methods, how robust and generalizable is the proposed model?
- (4) RQ4: How do different hyperparameter settings impact the model?

4.1. Datasets

Specifically, we used the following datasets in our tests:

Amazon-Book: A subset from the Amazon product review dataset, one of the most classic and widely used benchmark datasets in recommender systems, containing user ratings, reviews and timestamps for books.

Gowalla: A location-based social network dataset containing user check-ins (user ID, venue ID, timestamp), representing a typical implicit feedback scenario.

Tmall: Sourced from Alibaba’s Tmall platform, primarily comprising users shopping behaviors (clicks, add-to-cart, favorites, purchases) and corresponding timestamps. Table 3 provides detailed information about the datasets.

4.2. Evaluation metrics

This work uses two core evaluation metrics in recommender systems to assess top- p recommendation performance: Recall@ P and NDCG@ P . In this work, p takes values 20 and 40. The specific calculations for Recall and NDCG are as follows:

- (1) Recall: Measures the ability of the recommender system to identify items the user likes; that is, among all items the user likes, how many are successfully “retrieved” and included in the recommended list. It focuses on coverage and the ability to avoid misses. It can be defined as

$$\text{Recall} = \frac{\sum_{u \in U} R(u) \cap T(U)}{\sum_{u \in U} T(u)}. \quad (16)$$

Table 3. Detailed information of the datasets.

Dataset	#Users	#Items	#Interactions	Density
Gowalla	50 821	57 440	1 172 425	4.0×10^{-4}
Amazon	78 578	77 801	2 240 156	3.7×10^{-4}
Tmall	47 939	41 390	2 357 450	1.2×10^{-4}

In the formula, $R(u)$ represents the set of the top- p items recommended to user u ; $T(u)$ represents the set of items that user u truly likes (i.e. the real positive samples).

- (2) Normalized Discounted Cumulative Gain (NDCG): measures the ranking quality of the recommendation list. It cares not only about whether the recommended items are relevant, but also about their positions in the list. NDCG is a standardized value between 0 and 1; the closer to 1, the more ideal the ranking. Its calculation is as follows:

- (i) CG (Cumulative Gain): CG simply sums the relevance scores of the top p items in the recommendation list, without considering positional information. The relevance score can be represented as rel_i , where if the item is liked by the user $rel_i = 1$, otherwise 0.

$$CG_P = \sum_{i=1}^K \frac{rel_i - 1}{\log_2(i + 1)}. \quad (17)$$

- (ii) Discounted Cumulative Gain (DCG): DCG extends CG by introducing a position-based discount, so items appearing earlier in the ranking are more valuable than those appearing later. When evaluating the quality of a ranking list, relevant results at the top are more valuable than those at the bottom. The most commonly used formula is

$$DCG_p = \sum_{i=1}^P \frac{2^{rel_i} - 1}{\log_2(i + 1)}. \quad (18)$$

- (iii) Ideal Discounted Cumulative Gain (IDCG): IDCG is the maximum possible DCG achievable for a list. It is computed by taking all items relevant to the user, sorting them in decreasing order of relevance, and then computing the DCG of this ideal list.
- (v) Normalized Discounted Cumulative Gain (NDCG): NDCG is obtained by normalizing DCG by IDCG. By taking the ratio of DCG to IDCG, NDCG maps the ranking quality to the interval $[0, 1]$. A value of 1 indicates that the current ranking matches the ideal state perfectly; a value approaching 0 reflects a significant gap between the actual ranking and the optimum. This design allows NDCG to accurately measure how close the ranking results are to the ideal state and provides a unified standard for comparing the quality of different recommendation lists.

$$IDCG_p = \sum_{i=1}^{|REL|} \frac{2^{rel_i} - 1}{\log_2(i + 1)}, \quad (19)$$

$$NDCG_p = \frac{DCG_p}{IDCG_p}.$$

Table 4. Experimental results.

Method	Gowalla			Amazon-book			Tmall		
	R@20	R@40	N@20	N@40	R@20	R@40	N@20	R@40	N@40
NGCF	0.1404	0.2070	0.0817	0.0988	0.0534	0.0868	0.0387	0.0420	0.0364
LightGCN	0.1782	0.2579	0.1054	0.1253	0.0730	0.1149	0.0546	0.0557	0.0600
Mult-VAE	0.1793	0.2535	0.1079	0.1267	0.0736	0.1126	0.0559	0.0531	0.0473
CVGA	0.1822	0.2609	0.1126	0.1326	0.0849	0.1291	0.0660	0.0626	0.0560
HCCF	0.1819	0.2604	0.1061	0.1266	0.0845	0.1292	0.0627	0.0625	0.0550
LightGCL	0.1826	0.2600	0.1079	0.1285	0.0840	0.1279	0.0644	0.0634	0.0564
KPA-GCL	0.1862	0.2587	0.1098	0.1299	0.0764	0.1302	0.0603	0.0625	0.0524
DisneHAN	0.1438	0.2080	0.0829	0.0997	0.0543	0.0867	0.0405	0.0417	0.0378
DisneGCN	0.1380	0.2004	0.0795	0.0962	0.0483	0.0779	0.0354	0.0422	0.0380
DCCF	0.1870	0.2641	0.1120	0.1322	0.0888	0.1342	0.0650	0.0668	0.0598
DGCL	0.1790	0.2488	0.1090	0.1254	0.0688	0.1059	0.0509	0.0527	0.0482
DGCF	0.1783	0.2517	0.1074	0.1260	0.0690	0.1075	0.0516	0.0543	0.0486
AdaDCL	0.1887	0.2663	0.1128	0.1330	0.0902	0.1365	0.0693	0.0676	0.0601
BIGCF	0.2086	0.2883	0.1242	0.1450	0.0989	0.1468	0.0761	0.0755	0.0680
Ours	0.2097	0.2895	0.1260	0.1462	0.0996	0.1483	0.0778	0.0761	0.0691

4.3. Hyperparameter settings

To ensure fair experiments, this work adopts experimental settings consistent with mainstream research. We use Adam as the optimizer for the model. All models use the same embedding dimensionality of 64, a batch size of 4096, and Xavier initialization. For all baseline methods, we determine their best hyperparameters by following the optimal settings reported in their original papers and conducting grid search. For the CaDCR model, the number of intents K is chosen from [32, 64, 128, 256], and the number of GCN layers is explored in the range [1, 2, 3, 4].

4.4. Baseline comparison

We conduct a comprehensive comparison between the proposed CaDCR model and 14 state-of-the-art recommendation models in Recall@ P and NDCG@ P ($P = 20$ and 40). The results are shown in detail in Table 4. Additionally, to assess the efficiency of the model, we compare the training times of CaDCR with disentangled-based recommendation methods across three datasets, as illustrated in Fig. 2.

(1) GCN-based recommender models

NGCF¹⁵: Encodes collaborative signals by propagating embeddings on the user-item graph, capturing high-order connectivity for better recommendations.

LightGCN¹⁶: Simplifies GCN by keeping only neighborhood aggregation with linear propagation, improving efficiency without sacrificing accuracy.

(2) Generative recommender models

CVGA³⁷: Frames recommendation as graph generation via a variational graph autoencoder, modeling user behavior distributions for near-linear complexity.

Multi-VAE³⁸: Applies variational autoencoders to implicit feedback using multinomial likelihood and Bayesian inference, with regularization linked to the information bottleneck principle.

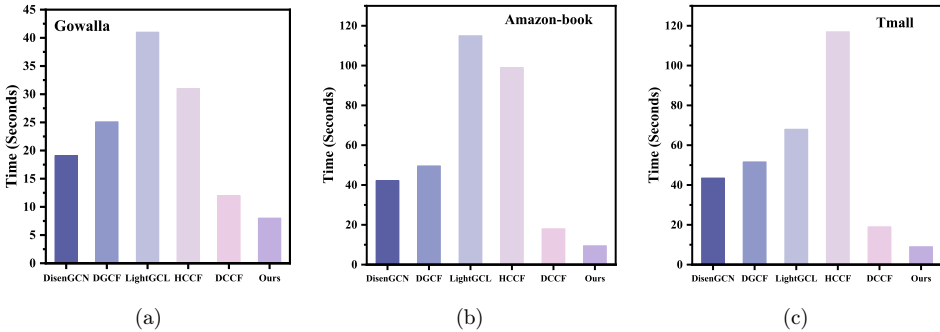


Fig. 2. Comparison of training time per epoch.

(3) Contrastive-learning-based recommender models

HCCF³⁹: Uses hypergraph contrastive learning to capture global user dependencies and improve robustness under sparsity.

LightGCL²¹: Employs SVD-based contrastive augmentation to preserve global relations and enhance robustness to sparsity and popularity bias.

KPA-GCL⁴⁰: Optimizes triple-embedding representations with relational graph convolutional networks, and uses contrastive learning to select high-quality sub-graphs, alleviating data sparsity and noise.

(4) Disentangled recommender models

DisenHAN⁴¹: Uses a heterogeneous graph attention network to decompose higher-order connections via meta-relations, iteratively identifying distinct feature facets for interpretable and effective representations.

DisenGCN⁶: Introduces a disentangled graph convolutional network with a dynamic neighborhood routing mechanism that assigns neighbors to different channels according to latent factors, achieving feature disentanglement and improved interpretability and robustness.

DCCF⁴²: Achieves intent-disentangled representation learning with a parameterized interaction mask generator to enable adaptive contrastive augmentation, capturing fine-grained latent factors and mitigating noise to boost recommendation robustness.

DGCL⁴³: Designs a disentangled graph contrastive learning approach that uses factorized representations and inter-factor contrastive learning to realize graph-level disentangled representations, improving downstream performance and model interpretability.

DGCF⁴⁴: Proposes a disentangled graph collaborative filtering model that dynamically models user intent distributions and optimizes an intent-aware interaction graph, producing independent and interpretable representations, substantially boosting performance and interpretability.

AdaDCL⁴⁵: Proposes adaptive disentangled contrastive learning with intent-aware modeling, layer-wise attention and dynamic negative sampling to mitigate over-smoothing and capture fine-grained user motivations.

BIGCF⁴⁶: Introduces a bidirectional-intent guided GCF model combining Gaussian graph generation and intent reconstruction, with nonaugmented GCR to reduce data sparsity effects.

4.5. Performance comparison

4.5.1. Overall performance analysis (Q1)

We validated our approach on several public datasets with varying characteristics. Experimental results show that, compared with baselines, our method maintains strong recall but also achieves improvements in ranking quality. This result confirms

that combining the causal inference process with machine learning representation methods significantly improves model design.

The effectiveness of our approach lies in two core innovations.

First, we introduce causal intervention to disentangle individual preferences from population effects — two factors traditionally conflated in user behavior data. Unlike conventional models that mix these signals in a single representation space, our structured causal model separates them, enabling accurate modeling of stable, long-term user interests. Second, we propose a dual-space contrastive learning mechanism that jointly regularizes the interaction of topology and intent representation spaces. This preserves local structural semantics while enforcing global distribution of uniformity, producing representations that are both discriminative and robust. Our approach avoids heuristic data augmentation by adaptively constructing contrastive samples, ensuring training stability and consistent representation of quality.

The intent-aware propagation mechanism enables effective capture of high-order collaborative signals, because it adaptively adjusts the information propagation paths according to different intent types. In the propagation of individual intents, the model emphasizes connections based on interest similarity; in the propagation of collective intents, it focuses on popularity-driven association patterns. This differentiated propagation strategy ensures multi-faceted modeling of complex user–item relationships, enriching the final feature representations.

4.5.2. Ablation analysis (Q2)

To validate the effectiveness of each core component of our system, we investigate the contributions of these modules: (i) Bidirectional Intent Modeling (BIM); (ii) GCR; (iii) Causal Intervention Mechanism (CaI). Table 5 details the ablation results

Table 5. Ablation study.

Method	R@20	N@20	R@40	N@40
Dataset: Gowalla				
BIM	0.2083	0.2885	0.1237	0.1442
GCR	0.2094	0.2876	0.1253	0.1446
Cal	0.2082	0.2872	0.1233	0.1453
Ours	0.2097	0.2895	0.1260	0.1462
Dataset: Amazon-book				
BIM	0.0992	0.1477	0.0771	0.0928
GCR	0.0987	0.1481	0.0768	0.0925
Cal	0.0992	0.1479	0.0768	0.0929
Ours	0.0996	0.1483	0.0778	0.0932
Dataset: Tmall				
BIM	0.0754	0.1149	0.0534	0.0686
GCR	0.0756	0.1133	0.0528	0.0682
Cal	0.0749	0.1148	0.0533	0.0683
Ours	0.0761	0.1173	0.0540	0.0691

on Gowalla, Amazon-book and Tmall datasets. The main results we obtained can be summarized as follows:

If we remove BIM, we report a noticeable drop in performance across all datasets, indicating that BIM can effectively separate individual user preferences from population effects, and, by decoupling these distinct motivational drivers, it substantially improves the model’s ability to capture true user interests.

The removal of GCR causes performance degradation on all datasets, demonstrating the value of graph-based contrastive regularization in enhancing representational discriminability. This module builds a contrastive learning task in the dual spaces, enriching self-supervised signals while ensuring representations preserve local structure and follow global distribution patterns, thus boosting generalization.

The removal of CaI leads to performance declines. This confirms the key role of causal intervention in mitigating confounding bias. Through this mechanism, the model effectively separates spurious correlations among intent factors, enabling more accurate estimation of true user preferences and improved recommendation accuracy.

4.5.3. Deeper analysis (Q3)

We thus investigated the model’s robustness to data sparsity and noise interference. The experimental results show that the proposed method exhibits advantages in addressing these two key challenges. We report in detail the results we obtained in our tests.

(1) Robustness to Sparsity

Users are divided into four groups based on their interaction frequency with items, and the performance of different methods is compared under sparse interaction scenarios. As shown in Fig. 3, as user interaction data decrease, all baselines experience varying degrees of performance decline; however, our method consistently maintains

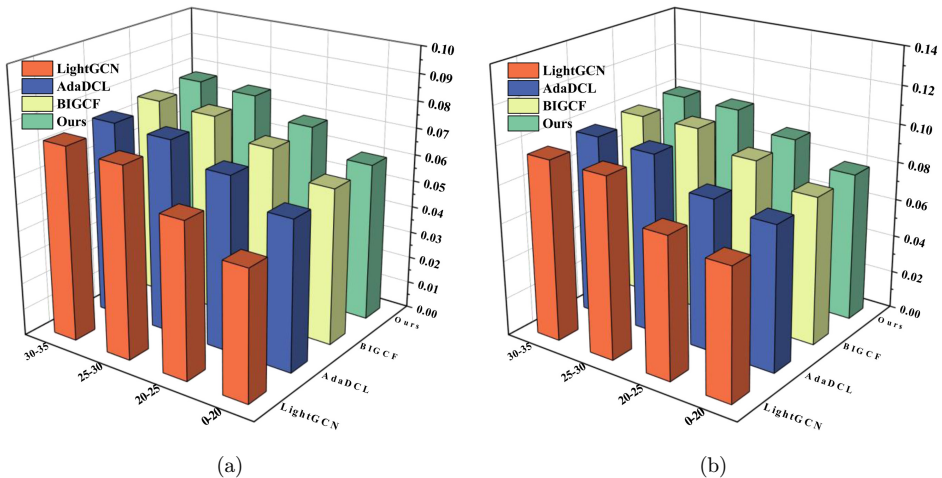


Fig. 3. Sparsity analysis of CaDCR.

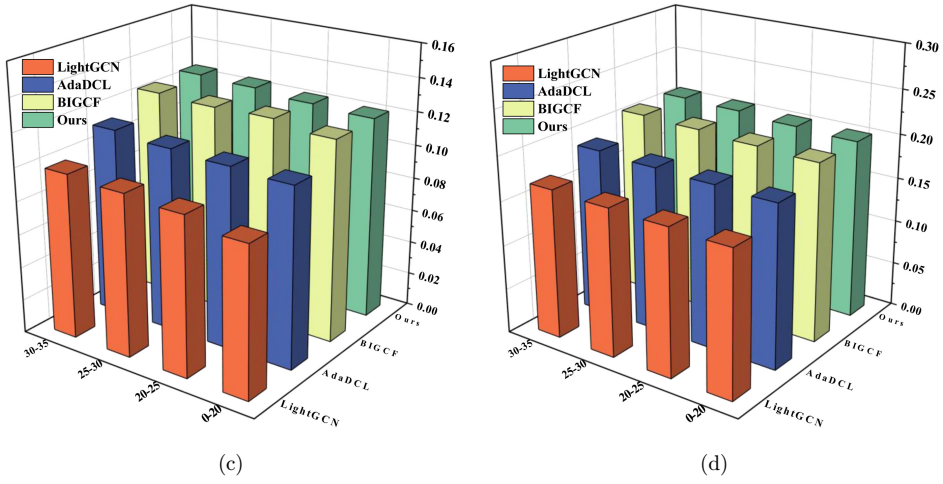


Fig. 3. (Continued)

the best performance. In the group with the scarcest interaction data, CaDCR demonstrates a larger performance advantage over other baseline models. This advantage mainly stems from the causal intervention mechanism, which effectively separates stable user preference features from limited observed data.

(2) Robustness to noise

We validated the CaDCR's ability to resist data noise by injecting noisy interactions data with varying proportions. Obtained results are reported in Fig. 4 and we point out that as the noise ratio increases, all baseline models exhibit performance drop, but our method shows the strongest resistance to interference. Even at high noise levels, in fact, CaDCR maintains relatively stable performance, with performance decline significantly smaller than that of other baselines. This phenomenon benefits from the dual-space contrastive learning mechanism in the model, which constructs a robust representation space enabling the model to effectively distinguish

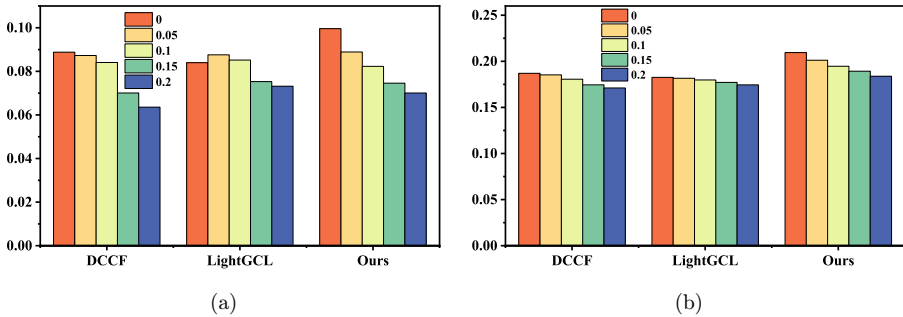


Fig. 4. Noise resistance analysis of CaDCR.

genuine user preferences from random noise. The two experiments about robustness analysis leads to our approach displaying high fault tolerance to data defects.

4.5.4. Hyperparameter analysis (Q_4)

K facilitates a deeper understanding of the behavioral characteristics of GNNs and multi-intent disentanglement in recommendation systems, while L enables the model to fully capture complex patterns while avoiding overfitting or structural redundancy. To this end, this section primarily analyzes these two key hyperparameters to explore how architectural parameters affect performance.

The results for parameter L are shown in Fig. 5(a). As the network depth increases, the model performance exhibits a rise-then-fall trend. Specifically, in the earlier, shallower stages, increasing L allows the model to learn richer high-order connectivity, thereby improving recommendation accuracy. However, as L continues to increase, performance declines significantly, mainly due to the over-smoothing issue that arises in overly deep GNNs. The results indicate that across all three datasets, $L = 3$ achieves a favorable balance, enabling the model to fully leverage deep information in the graph structure while avoiding the negative effects of over-smoothing.

The results for parameter K are shown in Fig. 5(b). When K is small, the model cannot adequately capture the diversity of user interests, leading to insufficient representational capacity. As K increases, the model can learn finer-grained intent decompositions, enabling a more precise characterization of personalized interests. However, when K becomes too large, performance improvements plateau or even decline, as overly fine intent partitioning introduces noise and redundancy and increases training difficulty. Based on our experiments, we conclude that K approximately equal to 128 is the optimal choice.

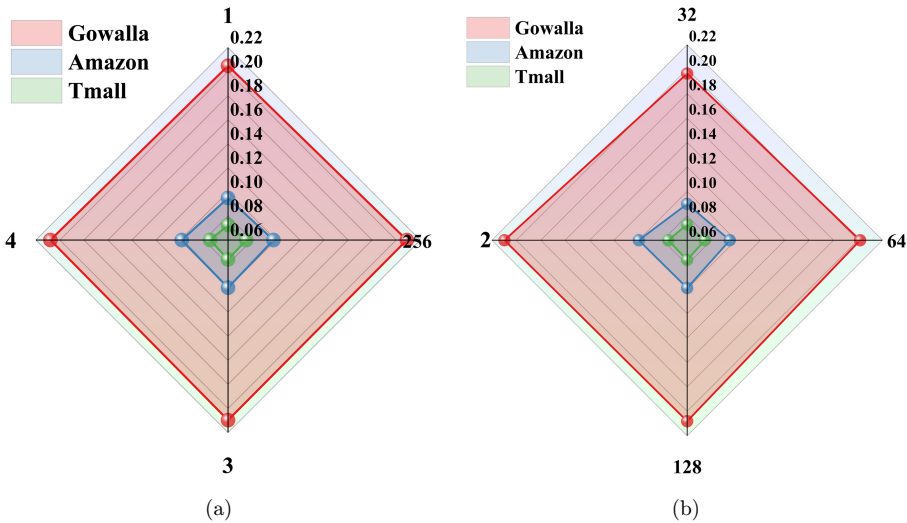


Fig. 5. Hyperparameter analysis of CaDCR (Recall@20).

5. Conclusions and Future Works

This paper tackles key issues in recommender systems, such as data sparsity, popularity bias and user-intent confusion, by proposing CaDCR, a causality-enhanced disentangled contrastive recommendation method. By constructing a causal disentanglement framework, it achieves effective separation of individual user intents and collective intents, learning unbiased representations that better reflect true user preferences. The dual-space contrastive regularization mechanism overcomes the limitations of traditional contrastive learning, preserving the semantic integrity of the graph structure while avoiding the uncertainties introduced by manually designed data augmentations. Through high-order cooperative signal encoding and intent-aware embedded propagation, the model can fully exploit deep information in the user-item interaction graph. Experiments on real datasets such as Gowalla, Amazon-book and Tmall show that CaDCR significantly outperforms existing mainstream methods in key metrics like Recall and NDCG, validating the effectiveness and superiority of the proposed approach.

The current approach still has certain limitations. For instance, the granularity of intent modeling remains relatively coarse, insufficiently capturing the micro-dynamics of user preferences. Furthermore, the model's generalization capability has not yet been validated in more complex scenarios such as cross-domain and sequential recommendation. Future work will explore more fine-grained intent modeling and extend the method to broader real-world scenarios, such as cross-domain and sequential recommendation, further improving the performance and practicality of recommender systems.

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